178054: foliated biotite granodiorite, Bell Chamber Well

Location and sampling

YOUANMI (SH 50-4), ATLEY (2741)
MGA Zone 50, 709810E 6889310N

Sampled on 29 October 2002

The sample was taken from the southern edge of a large rocky pavement 3 km northeast of Bell Chamber Well.

Tectonic unit/relations

The sample is from a dark-grey, medium- and even-grained, strongly foliated biotite granodiorite within the Youanmi Shear Zone, Southern Cross Granite–Greenstone Terrane, Yilgarn Craton (Chen, 2003). The foliation strikes northeast and dips steeply to the northwest. At the sampling site, the foliated granodiorite has been intruded by white pegmatite dykes up to 2 cm thick and aligned parallel to the foliation. The sample is free of pegmatite dykes and was collected to constrain the age of rocks within the Youanmi Shear Zone.

Petrographic description

The principal minerals are plagioclase (50–55 vol.%), K-feldspar (25 vol.%), biotite (2 vol.%), and opaque oxide (1 vol.%), with trace amounts of accessory titanite, magnetite, altered allanite, and zircon, and trace amounts of secondary chlorite, sericite, epidote, and leucoxene. This is a protomylonitic augen gneiss of granodioritic or monzogranitic composition. Augen of plagioclase up to 3 cm long are enclosed in a matrix composed of lamellae of plagioclase, quartz, and K-feldspar. The augen enclose minor biotite and comprise up to 20% of the thin section. The bulk of the rock is recrystallized, but there are rare larger grains of quartz up to 2 mm in length. Lamellae of fine-grained, strongly oriented biotite are abundant and define a layer-parallel schistosity, with lenses of quartz parallel to the schistosity. The c-axes of quartz in these lenses are typically at a low angle to the foliation. Recrystallized microcline and plagioclase are also abundant, mostly in separate layers that are best seen on the stained offcut because the grain size is too small (typically <50 µm) for adequate optical identification. Larger grains of microcline and plagioclase are up to 0.5 mm in diameter. Feldspathic aggregates contain minor myrmekite. Disseminated accessory magnetite, titanite, and apatite are up to 0.5 mm in grain size, and rare altered allanite is rimmed by epidote. Rare zircon crystals are about 40 to 100 µm. Cores of plagioclase are partly altered to sericite and clinozoisite. Some of the biotite has been altered to chlorite and leucoxene, with lenses and lamellae of metamorphic sericite parallel to the biotite lamellae. Very minor epidote, which is either granular or rims altered allanite, is probably metamorphic. Some titanite is partly altered to leucoxene. Metamorphism at about the greenschist to amphibolite facies transition was followed by weak low-temperature alteration.

Zircon morphology

The zircons isolated from this sample are typically colourless, straw-yellow, dark brown or black, euhedral and elongate, and between 50 × 80 µm and 160 × 80 µm in size. Most grains have faint internal zonation but some are internally structureless. Many grains have a speckled appearance in transmitted light, and many are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 24 April 2003. The counter deadtime during the analysis session was 24 ns. Nine analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 1.15% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 1.1, 9.1, 12.1, 13.1, and 14.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Fourteen analyses were obtained from 14 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with both ancient and recent episodes of radiogenic-Pb loss. Eleven concordant and highly discordant analyses of 11 zircons have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2674 ± 9 Ma (chi-squared = 0.73). The remaining analyses (4.1, 9.1, 13.1) are discordant and cannot be grouped.

The date of 2674 ± 9 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 11 concordant and highly discordant analyses of 11 zircons is interpreted as the age of igneous crystallization of the granodiorite. Discordant analyses 4.1, 9.1, and 13.1 are interpreted to be of sites that have undergone ancient radiogenic-Pb loss.

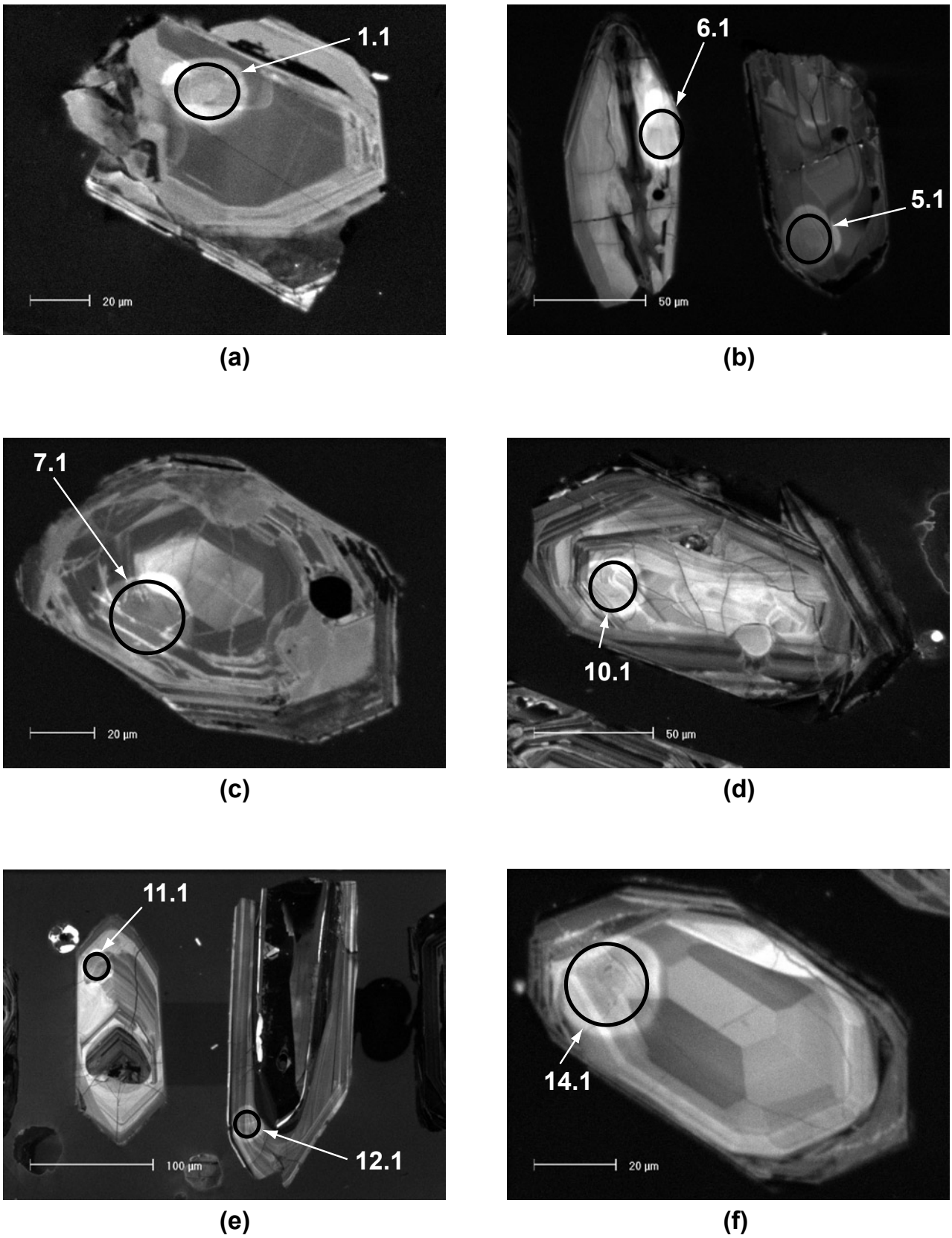

Figure 1. Cathodoluminescence images of representative zircons from sample 178054: foliated biotite granodiorite, Bell Chamber Well

Table 1. Ion microprobe analytical results for zircons from sample 178054: foliated biotite granodiorite, Bell Chamber Well

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	448	133	260	0.909	0.18214	0.00095	0.09827	0.00168	0.5099	0.0061	12.804	0.174	99	2 672	9
2.1	152	147	93	1.413	0.18108	0.00191	0.24068	0.00394	0.4777	0.0060	11.926	0.207	95	2 663	17
3.1	182	129	112	1.130	0.18306	0.00158	0.20906	0.00315	0.4969	0.0061	12.543	0.200	97	2 681	14
4.1	668	232	330	0.686	0.17430	0.00078	0.10072	0.00139	0.4394	0.0052	10.559	0.138	90	2 599	7
5.1	190	70	102	0.469	0.18158	0.00127	0.10123	0.00207	0.4801	0.0059	12.020	0.179	95	2 667	12
6.1	72	34	43	0.578	0.18717	0.00242	0.12048	0.00440	0.5133	0.0069	13.246	0.262	98	2 717	21
7.1	314	386	211	1.058	0.18175	0.00119	0.35152	0.00258	0.4952	0.0059	12.410	0.178	97	2 669	11
8.1	204	95	118	0.507	0.18217	0.00120	0.12382	0.00205	0.5083	0.0062	12.768	0.186	99	2 673	11
9.1	2 079	750	928	0.674	0.15706	0.00047	0.19969	0.00091	0.3737	0.0043	8.092	0.100	84	2 424	5
10.1	142	96	84	0.867	0.18474	0.00177	0.20015	0.00345	0.4849	0.0061	12.351	0.206	95	2 696	16
11.1	133	118	65	1.096	0.18189	0.00193	0.23878	0.00394	0.3895	0.0049	9.767	0.170	79	2 670	18
12.1	160	93	98	2.261	0.18111	0.00209	0.19704	0.00431	0.4794	0.0059	11.970	0.215	95	2 663	19
13.1	777	76	170	1.307	0.14703	0.00113	0.04377	0.00215	0.2039	0.0024	4.132	0.061	52	2 312	13
14.1	178	157	122	3.523	0.18162	0.00235	0.31212	0.00511	0.4795	0.0059	12.006	0.228	95	2 668	21

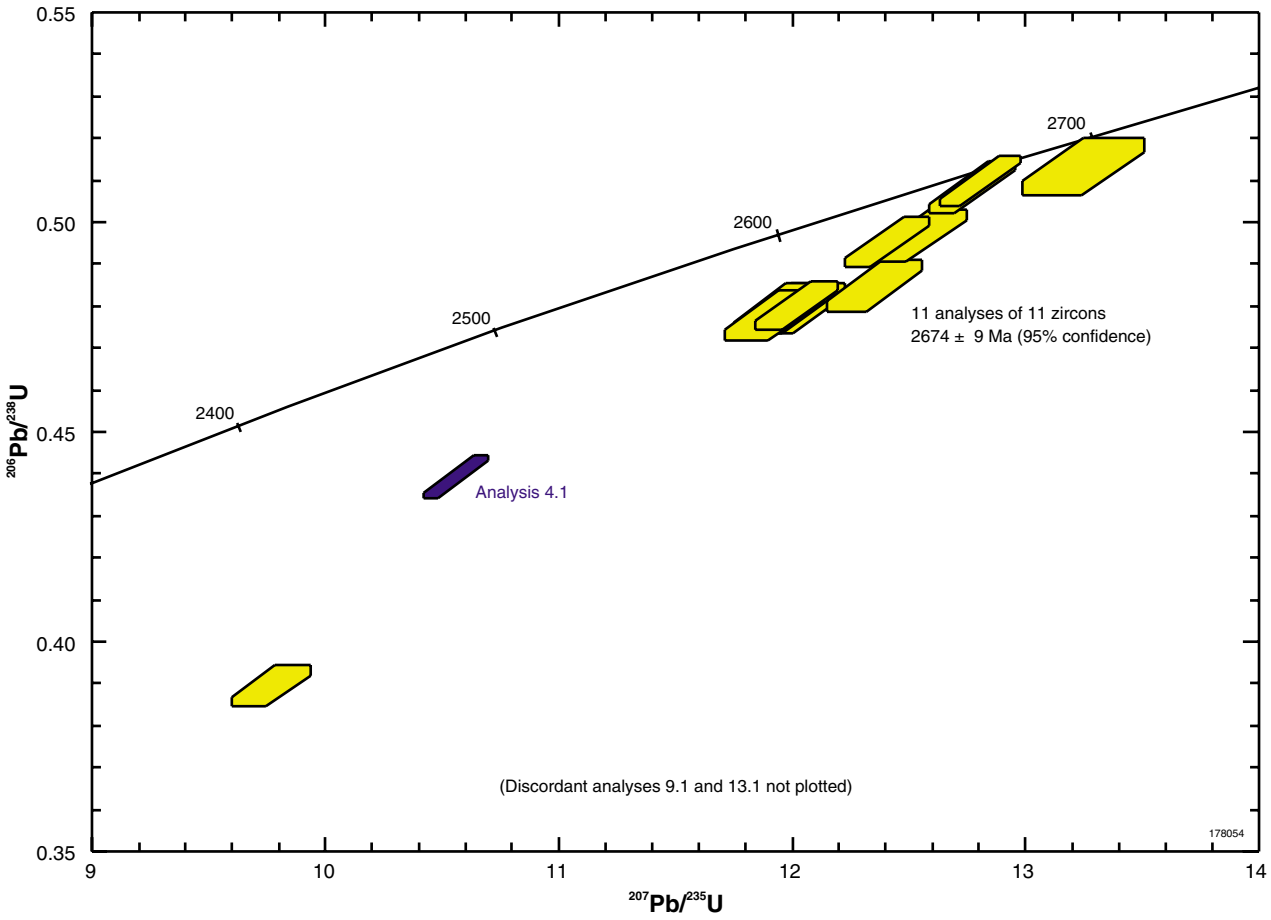

Figure 2. Concordia plot for sample 178054: foliated biotite granodiorite, Bell Chamber Well

Recommended reference for this publication:

NELSON, D. R., 2005, 178054: foliated biotite granodiorite, Bell Chamber Well; Geochronology dataset 582; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 24/04/2003; Data released: 30/06/2005